

CAP-CHAT, QC, Canada

WMO: 714280

Lat: 49.1092N

Lon: 66.6544W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-2.7	-14.4	2.5	-6.7	-9.9	3.2	-2.1	41.8	16.9	38.5	14.3	15.4	190	0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	73.0	65.0	70.0	63.2	67.7	61.8	66.5	71.5	64.5	69.0	62.7	66.8	11.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.1	69.5	62.3	84.0	67.3	60.6	79.1	65.6	31.2	71.7	29.7	69.2	28.3	66.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
35.8	31.2	27.8	DB	-12.2	81.9	5.5	4.4	-16.1	85.0	-19.4	87.6	-22.5	90.0	-26.5	93.2	
			WB	-12.9	70.9	5.9	2.3	-17.2	72.6	-20.6	73.9	-23.9	75.2	-28.2	76.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	14.5	15.1	23.3	34.6	45.1	54.2	60.8	59.9	52.9	42.8	33.5	23.0
	DBStd	17.55	10.05	9.38	8.79	5.86	5.67	5.58	4.48	4.70	5.79	5.92	7.41	8.38
	HDD50	5173	1102	977	827	463	177	19	0	1	34	238	497	838
	HDD65	9726	1567	1397	1292	912	619	325	145	170	363	688	945	1303
	CDD50	951	0	0	0	0	24	146	334	307	122	15	3	0
	CDD65	29	0	0	0	0	0	3	14	11	1	0	0	0
	CDH74	74	0	0	0	0	1	12	28	28	5	0	0	0
	CDH80	10	0	0	0	0	0	1	3	6	0	0	0	0
	WSAvg	11.9	14.3	13.1	12.9	11.8	9.5	9.4	10.1	10.4	11.2	12.6	13.4	14.5
	PrecAvg	33.9	2.5	2.2	2.4	2.2	3.0	3.0	3.2	3.1	3.1	3.7	3.3	3.0
Precipitation	PrecMax	39.4	4.6	3.9	4.9	4.6	5.5	7.5	6.7	6.7	9.1	6.4	7.5	4.7
	PrecMin	27.0	0.9	1.1	0.9	0.7	1.3	1.5	1.3	0.6	0.7	0.9	1.3	1.8
	PrecStd	3.3	1.1	0.8	1.1	1.0	1.2	1.2	1.3	1.6	1.7	1.4	1.5	0.7
	PrecStd	3.3	1.1	0.8	1.1	1.0	1.2	1.2	1.3	1.6	1.7	1.4	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	44.6	39.8	48.7	55.9	68.7	74.8	77.6	77.2	73.5	67.8	64.4	47.9	
	0.4% MCWB	41.2	37.0	44.1	45.6	56.7	63.5	67.2	67.3	65.1	60.3	58.0	44.0	
	2% DB	35.6	34.0	40.6	48.6	61.0	68.6	72.9	72.0	68.4	58.2	52.3	40.3	
	2% MCWB	33.3	32.1	36.5	42.1	52.6	60.4	65.3	64.9	61.6	53.1	46.8	37.4	
	5% DB	32.1	31.2	37.1	44.4	56.6	65.4	70.0	69.2	64.3	53.9	45.5	36.5	
	5% MCWB	30.8	30.0	33.9	39.7	49.9	58.5	63.7	63.5	59.1	50.0	41.9	34.6	
	10% DB	29.0	28.9	34.2	41.9	53.4	62.6	67.7	66.9	61.0	50.6	42.3	33.8	
	10% MCWB	27.6	27.6	31.8	37.9	48.0	56.7	62.2	62.0	56.8	47.3	39.7	32.4	
	0.4% WB	41.1	37.6	44.4	49.1	59.1	66.0	69.9	69.3	67.4	60.8	57.8	44.3	
	0.4% MCDB	44.0	39.9	48.2	53.7	66.9	72.9	74.9	74.4	72.5	65.2	64.9	47.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.9	32.5	37.4	42.7	53.9	62.1	66.6	66.2	62.9	53.4	47.8	38.1	
	2% MCDB	35.4	33.7	39.7	46.9	59.1	66.9	71.2	71.3	67.4	57.5	51.7	40.0	
	5% WB	30.8	30.2	34.3	40.3	51.0	59.7	64.6	64.2	59.7	50.0	42.6	34.8	
	5% MCDB	31.9	31.4	36.6	43.6	55.7	64.1	69.1	68.3	63.5	52.9	45.2	36.3	
	10% WB	27.4	27.9	32.4	38.5	48.5	57.6	62.8	62.4	57.4	47.4	40.2	32.7	
	10% MCDB	28.6	29.0	34.0	41.3	52.6	61.8	66.9	66.1	60.8	50.0	42.1	33.8	
Mean Daily Temperature Range	MDBR	13.0	12.7	12.2	10.5	12.0	12.3	11.6	11.2	11.7	10.2	9.8	10.4	
	5% DB	17.5	13.3	14.0	15.5	17.6	16.4	15.2	14.3	16.6	15.7	15.7	15.0	
	MCWBR	17.2	12.1	12.0	11.5	11.8	10.7	10.1	9.8	12.1	12.2	13.1	13.9	
	5% WB	17.7	13.1	13.2	14.4	16.1	15.2	14.0	13.3	15.8	14.9	15.0	14.5	
	MCWBR	17.0	12.2	11.8	11.2	11.7	10.9	10.3	9.7	12.2	12.3	12.9	13.5	
	MCWBR	17.0	12.2	11.8	11.2	11.7	10.9	10.3	9.7	12.2	12.3	12.9	13.5	
Clear-Sky Solar Irradiance	taub	0.269	0.278	0.306	0.385	0.365	0.385	0.419	0.388	0.363	0.336	0.296	0.277	
	taud	2.360	2.306	2.303	2.218	2.396	2.371	2.313	2.403	2.450	2.500	2.520	2.398	
	Ebn at Noon	253	278	288	272	284	278	266	270	266	255	244	234	
	Edh at Noon	21	28	34	41	36	37	39	34	29	23	18	18	
All-Sky Solar Radiation	RadAvg	372	670	1064	1407	1649	1774	1715	1475	1098	654	340	255	
	RadStd	43	69	71	123	109	124	173	90	84	54	41	25	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	+2.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page